1. Six weak points of EC index calculus, measured on Weil-descent systems

Run 20261102T1639Z (frozen arms A2, A3, Y, P) Amendment-1 arm Y2, and Amendment-2/3/3a arm L (large primes, seed 20261104, run 2026-11-02 19:16–20:21Z) · **tool** `research/ic_weakpoints_20261102/tool` (Rust, std only), commits `1a572f0607` (arms) and `97b0965fa7` (diagnostics, summaries) · **protocol** `PROTOCOL.md` frozen before data · **host** one 4-vCPU cloud container, no isolation (exact counts only, no timing) · **seeds** base 20261102 (Y2: 20261103), derived per arm.

Scope. Gaudry/Diem descent: $E : y^2 = x^3 + ax + b$ over $\mathbb{F}_{p^n}$, random a, b, factor base $\{P : x(P) \in \mathbb{F}_p\}$, $n \in \{2, 3\}$. Not an ecbench session, not a leaderboard row, not admitted by the validation gate.

1.1. Outcomes

Row 1 solving degree	$n = 3$: degree 10 in 88/88 systems for $p = 31 \ldots 2^{30}$; Macaulay matrix 252×286, $c(3) = 1.03$–1.07×10^6 $\mathbb{F}_p$ mults (dense upper bound). $n = 2$: degree 3, 87 mults. Flat in p.
Row 5 geometry	Zero-dimensional 176/176, quotient dimension 64 = Bézout (87/88). $p \le 127$: 32 rational solutions, 5 genuine (equal to the independent point-arithmetic count), 27 non-split (84 % spurious), 2 singular, 0 positive-dimensional.
Row 3 yield	mean/λ = 0.88–1.19 over 20 curves; dispersion CI contains 1 in 19/20; heavy tails only from subfield targets $x_R \in \mathbb{F}_p$ (replayed: 25 decompositions, all $R + P - P$).
Row 4 rank	log Q correct 15/15; rank shortfall 0–3 (one curve 6); cofactor and duplicate-row controls pass 15/15; sign-drop bug detected 12/15.
Row 6 cost	IC/rho (negation-map reference) $w = 0$: 0.29–3.4; $w = 0.1$: 2.8×10^4–3.2×10^5; $w = 1$: 1.5×10^5–1.7×10^6. Rho ahead 7, unresolved 8, IC ahead 0.
FB size / LP	One-large-prime yield matches $3\varphi^2(1 - \varphi)$ (0.93–1.04); full-relation deficit in Y (0.85 at $\varphi = \frac{1}{2}$) not reproduced in Y2 (1.08), recorded as noise.
Derived	No large primes: best φ gives IC/rho plateau 684 ($w = 0.1$), 3761 ($w = 1$) for groups $\ge 2^{60}$, since both scale as $p^{\frac{3}{2}}$.
Large primes (arm L)	$n = 3$, $p = 127 \ldots 2039$, 2 curves each. Trials to a square system: 2LP at $r = \frac{2}{3}$ grows as $p^{1.27 \pm 0.04}$ (theory $\frac{4}{3}$, rho 1.5; prediction ≤ 1.4 holds); 0LP follows $3 - 2r$ where measurable ($r = 1$: 1.02 ± 0.03). log Q correct in 105/105 reached cells. Prediction (d) fails narrowly in 1/25 cells (2LP row weight 3.02× 0LP). Run 1 used a fixed +20 row margin that biased exponents low (2LP 1.12); refitted under Amendment 3, replay identical in every non-counter field.
Derived (2LP)	Extrapolated from $p \le 2^{11}$: IC/rho at $w = 0.1$ is 58 (2^{96}), 7.7 (2^{132}), below 1 near 2^{180}; at $w = 1$: 319, 42, 5.3 and above 1 to 2^{240}. Not ranked by §5a anywhere up to 2^{240}.

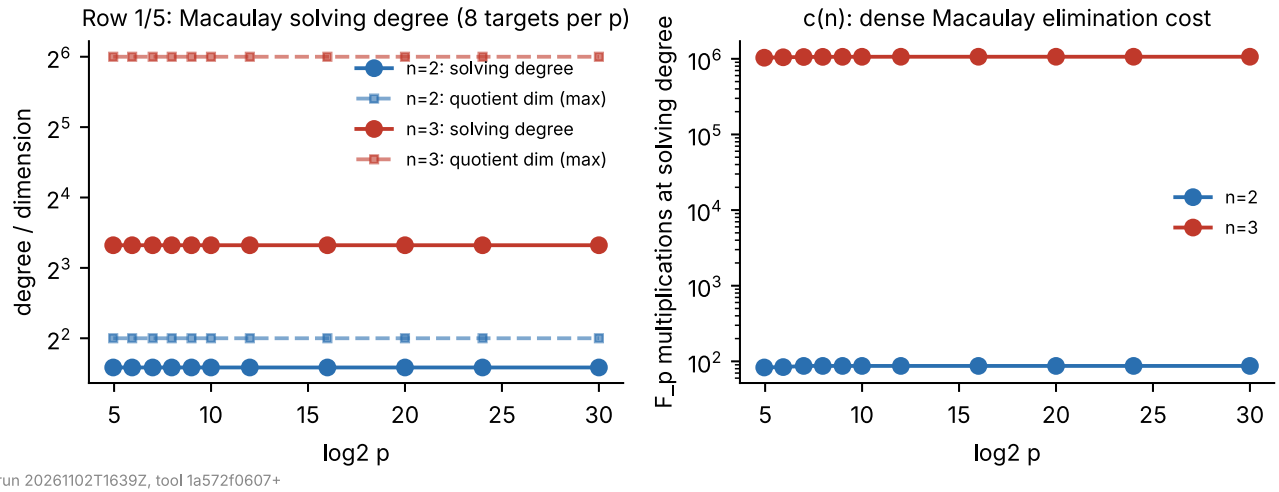


Figure 1: Rows 1 and 5. Solving degree, quotient dimension and dense Macaulay cost at the solving degree against $\log_2 p$; 8 random curves and targets per p; band = min–max. Run 20261102T1639Z, `data/algebra_n{2,3}.jsonl`.

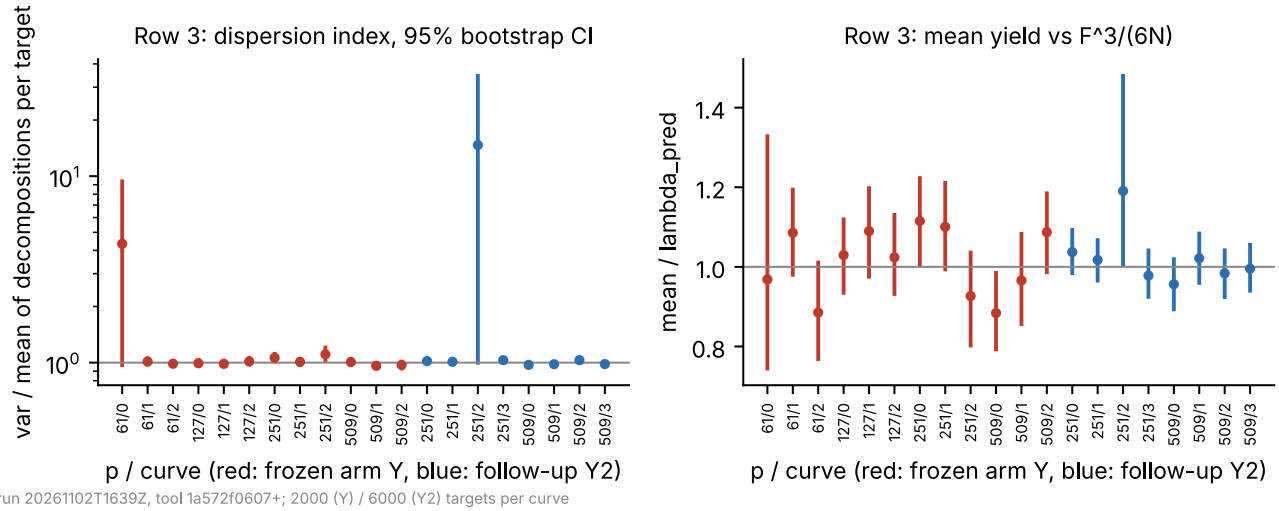


Figure 2: Row 3. Dispersion index and mean/λ per curve with 95 % bootstrap CIs (1000 resamples, computed natively); 2000 (Y) or 6000 (Y2) targets per curve. The two wide intervals come from single subfield targets. `data/yield*.jsonl`.

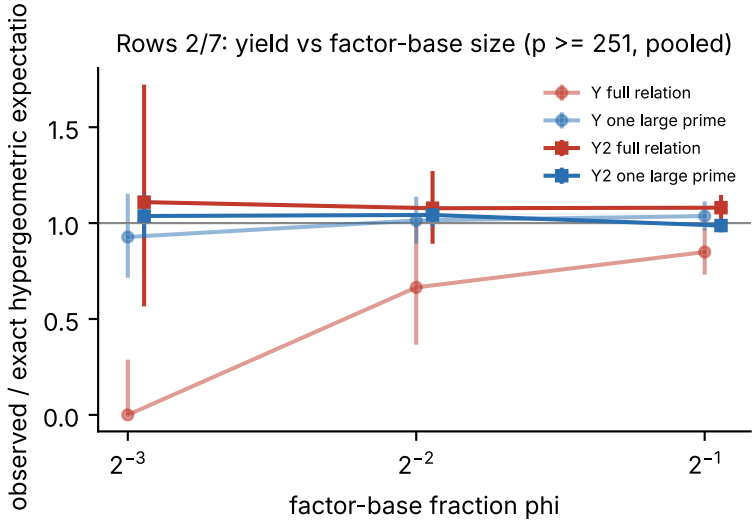


Figure 3: Rows 2/7. Observed over exact hypergeometric expectation for relations inside the first φ fraction of the factor base, pooled over $p \in \{251, 509\}$, Poisson 95 % CI. Y: 6 curves × 2000 targets; Y2: 8 curves × 6000.

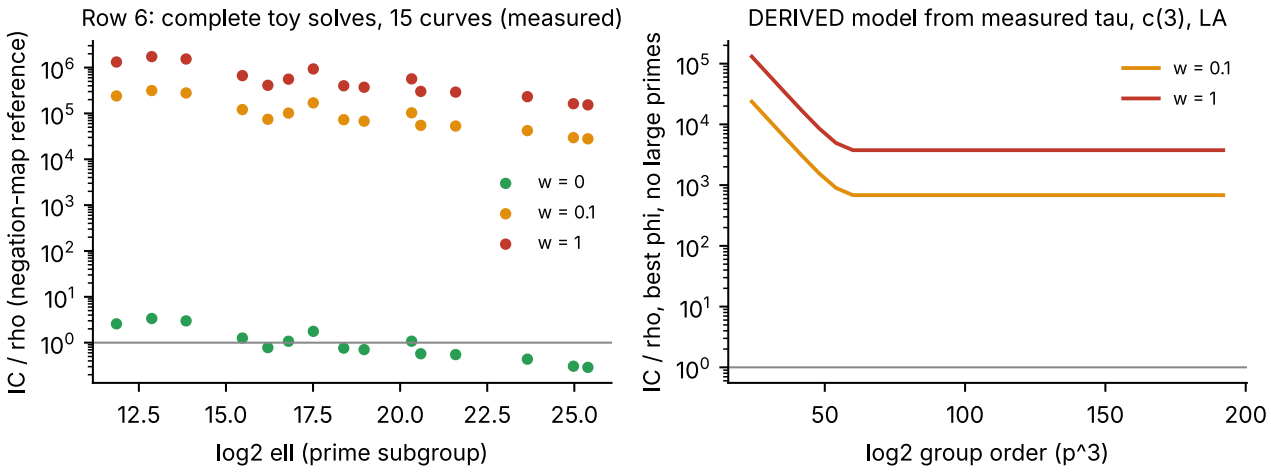


Figure 4: Row 6. Left: measured complete solves (15 curves, log Q verified), IC cost = walk additions + w (trials × $c(3)$ + linear algebra), against analytic negation-map rho with comparisons priced at w. Right: derived model from the measured constants; not a measurement.

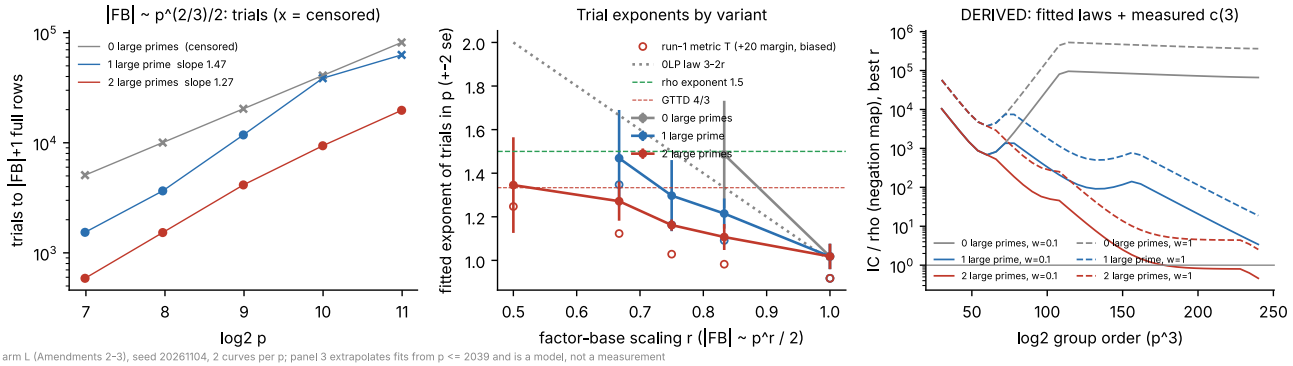


Figure 5: Large-prime variation (arm L). Left: trials until $|\mathrm{FB}| + 1$ full rows at $r = \frac{2}{3}$, geometric mean of 2 curves; × = censored at $40p$. Middle: fitted exponents ± 2 se by r (filled: square-system count; hollow: run-1 count with the +20 margin, biased low). Right: DERIVED IC/rho at the best measured r, extrapolating the fits by up to 70 bits; the 0LP curve is limited by the r grid. `data/dlp_p*.jsonl`, `data/summary_dlp.jsonl`.

1.2. Limits

Dense Macaulay elimination overstates $c(n)$ relative to F4; the decomposition oracle used for relation finding is point arithmetic (its additions are excluded and replaced by the Gröbner price); $n = 4$ was not measured; large-prime exponents come from $p \leq 2039$ and the 2LP slopes at $r \geq \frac{3}{4}$ (1.11–1.16) look pre-asymptotic, so the derived crossover is optimistic; toy sizes only; one host, no isolation, which is immaterial for exact counts but rules out any timing statement.